

Please Grab a handout in the back.

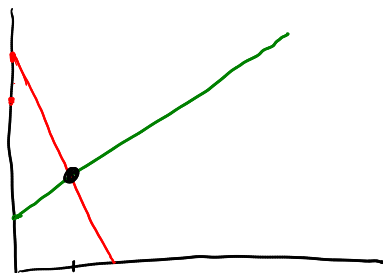
Today

Solving Nonlinear Function
Bracketing Methods: Most important: Bisection

Announcements / Reminders

Lab 1 due Thursday - Check Auto-graded!
HW 1 Released (Graded)

Linear

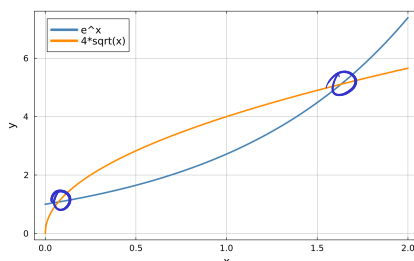


$$-2x + 4 = x + 1$$

$$x_2 = -2x_1 + 4 \quad x_2 = x_1 + 1$$

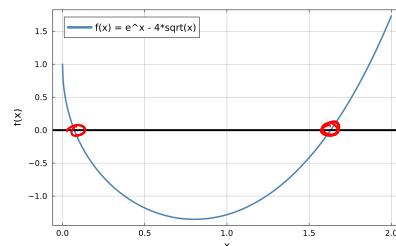
$$x = 1$$

Nonlinear



$$e^x = 4\sqrt{x}$$

no analytical in "simple functions"



$$f(x) = 0 = e^x - 4\sqrt{x}$$

find "roots"
x-values where $f(x) = 0$

Bracketing

Open

↑ Today

Incremental Search

Incremental Search

input : function f , search range $[x_{\min}, x_{\max}]$, number of steps n

output: list of brackets $[x_l, x_u]$, each containing a root

$$\Delta x \leftarrow (x_{\max} - x_{\min}) / n$$

brackets $\leftarrow \{\}$

$$x_l \leftarrow x_{\min}$$

for $k \leftarrow 1$ to n do

$$x_u \leftarrow x_l + \Delta x$$

if $f(x_l) \cdot f(x_u) < 0$ then

// sign change $\Rightarrow$ a root lies in $[x_l, x_u]$

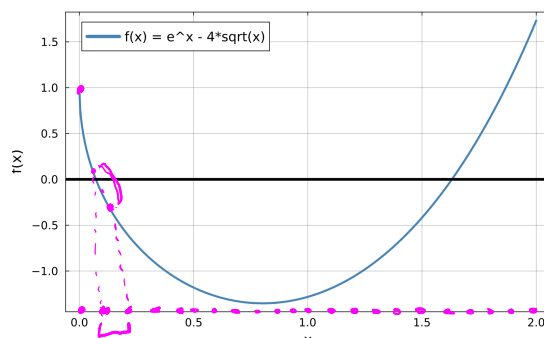
append $[x_l, x_u]$ to brackets

end

$$x_l \leftarrow x_u$$

end

return brackets



↑ if $f(x)$ is continuous and $f(x_l)$ and $f(x_r)$ have different signs, a root is between x_l and x_r

Strength: Can find multiple roots

Weakness: Can miss roots, slow at finding one root

(we will see shortly)

error $\rightarrow E_a = \frac{\Delta x}{n-1}$

Bisection

Bisection

input : function f , bracket $[x_l, x_u]$ with $f(x_l) \cdot f(x_u) < 0$,

tolerance ε_s

output: estimate x_r of the root

$$x_r \leftarrow x_l$$

repeat

$$x_{r,old} \leftarrow x_r$$

$$x_r \leftarrow (x_l + x_u) / 2$$

// keep the half that still brackets the root

if $f(x_l) \cdot f(x_r) < 0$ then

$$x_u \leftarrow x_r$$

else if $f(x_l) \cdot f(x_r) > 0$ then

$$x_l \leftarrow x_r$$

else

return x_r

end

$$\varepsilon_a \leftarrow \frac{|x_r - x_{r,old}|}{x_r} \cdot 100\%$$

until $\varepsilon_a < \varepsilon_s$

return x_r

// exact root

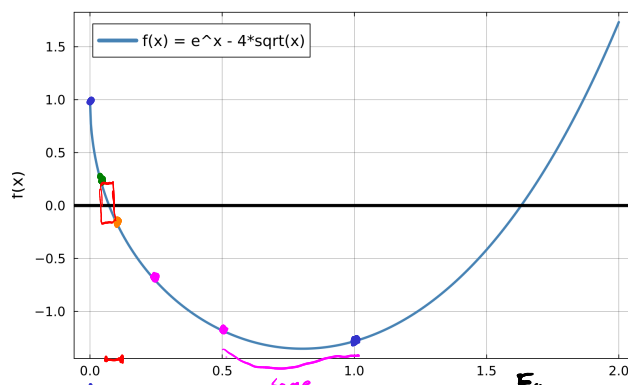
Iter: 1

2

3

4

5



$$H \text{ TRA } 0.0675, 0.125$$

E_a

$$\Delta x = 1$$

$$\Delta x = 0.5$$

$$\Delta x = 0.25$$

$$\Delta x = 0.125$$

$$\Delta x = 0.0625$$

$$\Delta x = 0.03125$$

$$E_a = \frac{\Delta x}{2^n}$$

$$E_a = \frac{\Delta x}{2^n}$$

function calls
for desired $\frac{\Delta x}{E_a}$

Incremental

$$E_a = \frac{\Delta x}{n-1}$$

$$n-1 = \frac{\Delta x}{E_a}$$

$$n = \frac{\Delta x}{E_a} + 1$$

$$O\left(\frac{\Delta x}{E_a}\right)$$

Bisection

$$E_a = \frac{\Delta x}{2^n}$$

$$2^n = \frac{\Delta x}{E_a}$$

$$n = \log_2\left(\frac{\Delta x}{E_a}\right)$$

$$O\left(\log\left(\frac{\Delta x}{E_a}\right)\right)$$

$$E_a = 0.01 \Delta x$$

$$\frac{\Delta x}{E_a} = 100$$

101 func
evals

$$\log(100) = 6.64$$

7 function
evals !

$$E_a = 0.0001 \Delta x$$

$$\frac{\Delta x}{E_a} = 10,000$$

10,001 func
evals

14 func
evals

False Position

False Position

input : function f , bracket $[x_l, x_u]$ with $f(x_l) \cdot f(x_u) < 0$,
tolerance ε_a

output: estimate x_r of the root

$x_r \leftarrow x_l$

repeat

$x_{r,old} \leftarrow x_r$

$$x_r \leftarrow x_u - \frac{f(x_u)(x_l - x_u)}{f(x_u) - f(x_l)}$$

// keep the side that still brackets the root

if $f(x_l) \cdot f(x_r) < 0$ then

| $x_u \leftarrow x_r$

else if $f(x_l) \cdot f(x_r) > 0$ then

| $x_l \leftarrow x_r$

else

| return x_r

end

$$\varepsilon_a \leftarrow \left| \frac{x_r - x_{r,old}}{x_r} \right| \cdot 100\%$$

until $\varepsilon_a < \varepsilon_a$

return x_r

← only difference
from bisection

Problem: Doesn't
handle
curvature well

